

Denis Khilkov

Fullstack Developer

Contact Information

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Professional Summary

Pragmatic Fullstack Developer with 5+ years building business-critical software — from web services in Go and Node.js to cross-platform desktop apps on Electron — with a strong backend focus and hands-on frontend experience in React and Vue. Proven track record delivering customer-facing portals and B2B integrations end-to-end: REST APIs on Fiber/Echo with GORM and PostgreSQL, JWT authentication, payment and mortgage integrations, internal CRM/ERP document workflows, and SPA-to-SSR migrations. Most recently focused on desktop application development and test automation — building a custom E2E testing framework on Playwright, plus soak/stress and memory-leak testing for unattended, always-on software. Experienced in evolving monolithic services into microservices on gRPC, hardening reliability with RabbitMQ-backed queues, and containerizing the full stack with Docker and Nginx. Remote-first, available for full-time roles, ready to own product features from API design to production delivery.

Key Achievements

- **E2E Test Automation Framework:** At TBFF, designed a custom scenario-driven end-to-end testing framework for Electron desktop apps (Playwright + TypeScript) — declarative scenario configs, a runner driving a real app instance across soak/stress cycles, per-step attribution of console errors, exceptions and crashes, and video reports with a virtual-cursor overlay.
- **Reliability of Always-On Software:** Diagnosed and eliminated renderer memory growth over 18+ hour unattended kiosk sessions and fixed leaking event listeners/timers, hardening the apps for continuous multi-day operation.
- **Customer Portal at Scale:** Built and grew the customer account portal at DOGMA from a Go MVP into a feature-rich product covering apartment search, paid online reservations integrated with an online cash register and fiscalization provider, and online mortgage origination integrated with multiple banking APIs.
- **Monolith → Microservices:** Refactored the customer portal monolith into focused Go microservices (deals service, upstream-CRM mirror service) communicating over gRPC, lifting reliability and enabling independent deployments.
- **Resilience via Messaging:** Introduced RabbitMQ as the backbone for talking to internal CRM, internal ERP, and third-party APIs — eliminating data-loss scenarios on outages and decoupling external integrations from request paths.
- **Data Migration:** Designed and ran an automated migration of the apartment catalog from a legacy CRM into PostgreSQL, enabling fast filtered search and unblocking modern UX features.
- **API Tooling:** Stood up a documented REST API on Fiber with Swaggo-generated OpenAPI specs, JWT auth/refresh middleware, GORM-backed PostgreSQL access, and Gothenberg-driven PDF generation; consolidated this stack as the template for new internal services.

Professional Experience

TBFF (Talking Birds & Flying Fish) — Fullstack Developer — Dec 2025 — May 2026

Experiential studio building interactive touchscreen kiosk installations for large-scale industry events

- Delivered cross-platform desktop applications (Electron, React, TypeScript, Vite) for dual-display touchscreen kiosks running unattended in kiosk mode: multi-screen narrative flows, timed mini-games, multilingual content, voice-over/audio, and auto-update.
- **Test automation:** designed a custom scenario-driven E2E testing framework on Playwright + Electron + TypeScript — declarative scenario configs describe user flows, a runner drives a real Electron instance across configurable soak/stress cycles, and an error-collection layer attributes console errors, exceptions and renderer crashes to the exact failing step.
- Made test runs reproducible and reviewable: video capture with a virtual-cursor overlay (desktopCapturer + ffmpeg WebM→MP4) and centralized data-testid selectors with a locator-resolution layer (testid / role / text / css) for specs stable against UI change.
- Built a second Playwright E2E suite for another kiosk app, covering app startup, slide transitions, inactivity/idle behavior, survey flows and token collection.
- **Memory-leak & reliability testing:** diagnosed renderer memory growth over 18+ hour kiosk sessions (soft navigation kept the renderer alive) and moved to a full app reload behind a seamless blackout transition; audited components and fixed a critical video event-listener leak.
- Authored and ran manual QA test plans covering dual-display behavior, full screen sequences, mini-games and the settings/kiosk configuration, complementing the automated suites.

DOGMA — Backend Developer — Sep 2019 — Jan 2024

Real-estate developer; in-house portfolio of customer, agency and counterparty portals integrated with the company's internal CRM and ERP systems

- Started full-stack on Node.js / Express with React and Vue: redesigned company landing pages and the main site, migrated from a SPA to SSR for faster first paint and SEO, and moved hosting from static web-hosting to Dockerized services behind Nginx.
- Transitioned fully to Go backend: launched the company's first Go service — an apartment catalog with filtered search — and ran an automated migration of apartment data from the legacy CRM into PostgreSQL.
- Built the customer account portal in Go: JWT authentication, SMS-code registration via a third-party SMS gateway, favorites, and live status of the customer's deals from the upstream CRM.
- Delivered paid online apartment reservations (4-day hold) integrated with an online cash register API and a fiscal-data provider for receipts.
- Implemented online mortgage origination inside the customer portal, integrated with multiple banking APIs covering different lenders.
- Built the counterparty portal: deal status, and electronic document workflow with the company's internal ERP system for counterparty accreditation.
- Designed and shipped the realtor-agency portal: REST API initially on Echo, later migrated to Fiber for throughput; OpenAPI documentation via Swaggo; JWT auth-token + refresh-token middleware; PostgreSQL access through GORM; PDF generation via Gothenberg.
- Released a translation service used by mobile and frontend teams across the company.
- Adopted RabbitMQ for asynchronous communication with internal CRM, internal ERP, and third-party APIs, improving fault tolerance for bursty and unreliable upstreams.
- Refactored the customer portal monolith into Go microservices (deals service, upstream-CRM mirror service) communicating over gRPC.

- Onboarded one frontend engineer and one backend engineer; transferred ownership of legacy areas.

ShtrafovNet.ru — Backend Developer — Feb 2024 — Apr 2024

Service for tracking and resolving traffic fines

- Built and refactored the REST API powering the public website and the web application.
- Streamlined endpoint contracts and internal data flows for faster, more predictable response times.

Technical Skills

Programming Languages: Go, JavaScript, TypeScript **Backend:** Go (Fiber, Echo, GORM), Node.js, Express, JWT (auth-token + refresh-token), OpenAPI/Swagger via Swaggo, Gothenberg (PDF generation), gRPC **Frontend & Desktop:** React, Vue, HTML, CSS, Electron (electron-vite, electron-builder), Vite **Testing & QA:** Playwright (E2E automation), Electron app testing, scenario / soak / stress testing, memory-leak diagnosis, manual test planning **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis **Messaging & RPC:** RabbitMQ, gRPC, REST API **APIs & Integrations:** internal CRM, internal ERP (accreditation, document workflow), banking APIs, online cash register / fiscal-data providers, SMS gateways, generic third-party REST APIs **DevOps & Tooling:** Docker, Nginx, Linux, Git, SSR migration

Education

KAIT 20 — College of Automation and Information Technologies — 2015 Industrial Automation Systems

Languages

- Russian — Native
- English — B1 (Intermediate)

Additional Information

- **Availability:** Immediately available for Backend / Fullstack Developer roles at product companies.
- **Work Preferences:** Remote-first, full-time. Not available for business trips.
- **Target Stack:** Go services, REST/gRPC APIs, PostgreSQL/Redis, RabbitMQ, Docker, with frontend on React or Vue when needed.

[Full CV.pdf](#)